

Incentives with Taxes and Transfers

Economic Impact Analysis of Government Measures



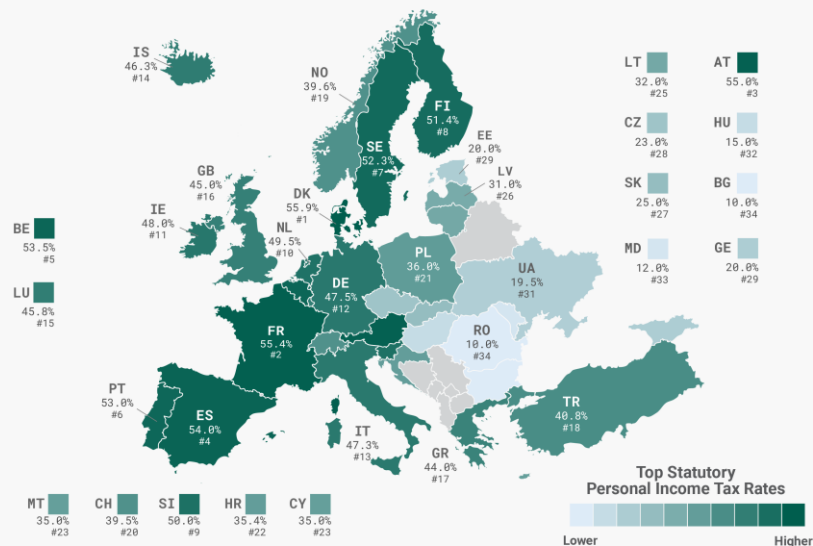
Gruber, Kapitel 20 (3) & Folien



Difference?

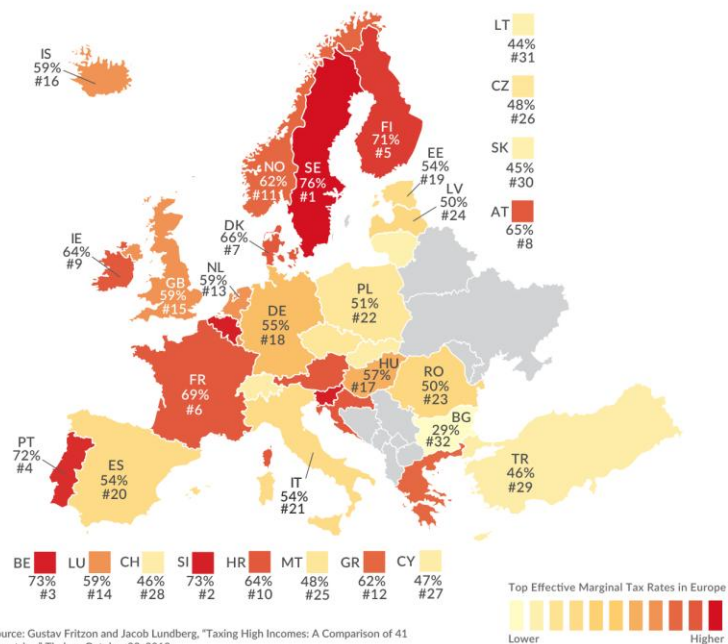
Top Personal Income Tax Rates in Europe

Top Statutory Personal Income Tax Rates in 36 Major European Countries, 2024



Note: Combined central and sub-central top personal income tax rates and surtaxes are shown. Social security contributions are not captured.
Source: European Commission, "Taxes in Europe Database v3," and PwC, "Worldwide Tax Summaries - Personal Income Tax (PIT) rates."

Top Effective Marginal Tax Rates in Europe



Source: Gustav Fritzon and Jacob Lundberg, "Taxing High Incomes: A Comparison of 41 Countries," Timbro, October 23, 2019.

Marginal Tax Rate & Effective Definition?

Marginal Tax Rate:

- The percentage of tax burden on the last Euro earned.
- **Example:** Someone who earns over €50,000 and pays 40% tax on every additional Euro has a marginal tax rate of 40%.

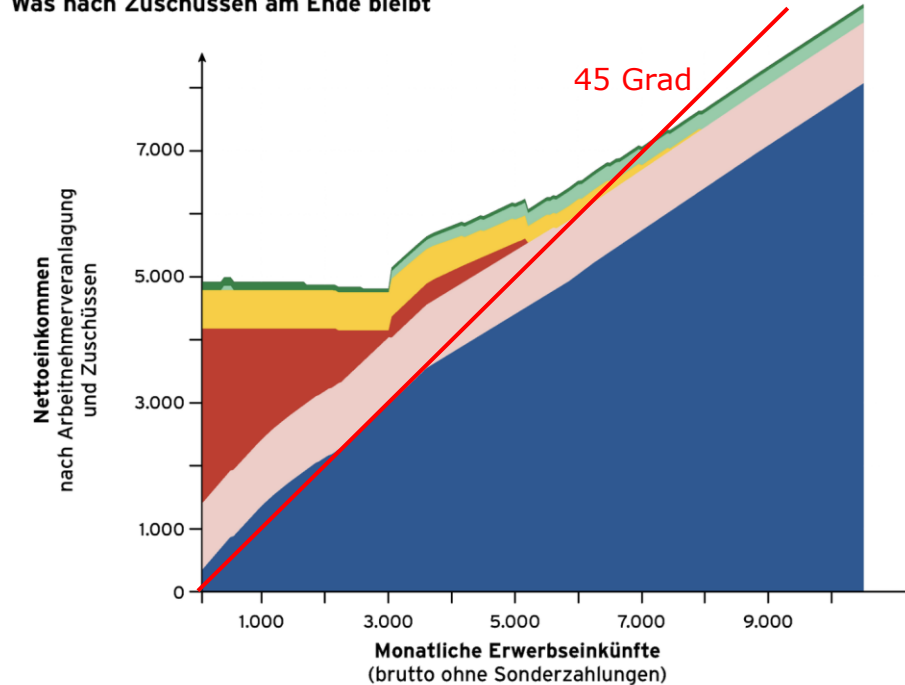
Effective Marginal Tax Rate:

- The actual burden on every additionally earned Euro—including taxes, social security contributions, and the loss of social benefits.
- **Example:** Someone who loses €600 in taxes and cancelled benefits for a €1,000 increase in income has an effective marginal tax rate of 60%.

Does Work Still Pay Off? Taxes and Transfers

4children,Household, Graz

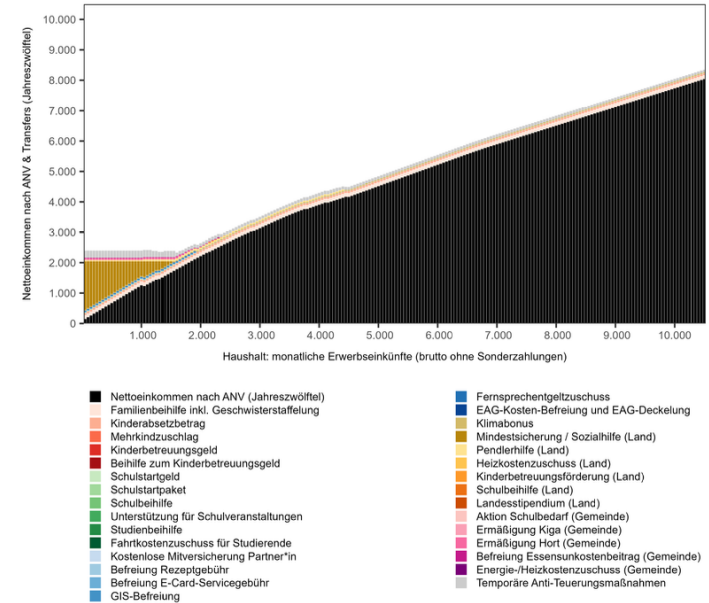
Was nach Zuschüssen am Ende bleibt



- U.a. Rezeptgebühr-, E-Card-Servicegebühr-, GIS-Befreiung, Klimabonus
- Kostenlose Mitversicherung
- U.a. Schulbeihilfe, Hortermäßigung, Feriencampzuschuss
- Mindestsicherung/Sozialhilfe, Wohnunterstützung, Heizkostenzuschuss, SozialCard
- Familienbeihilfe, Kinderabsetzbetrag, Mehrkindzuschlag, Schulstartgeld, Schulstartpaket
- Nettoeinkommen nach Arbeitnehmerveranlagung

[Link zu Artikel](#)

Conservative Example:
1-year-old child (7 years old),
Income distribution 50/50



[Link zu Studie](#)

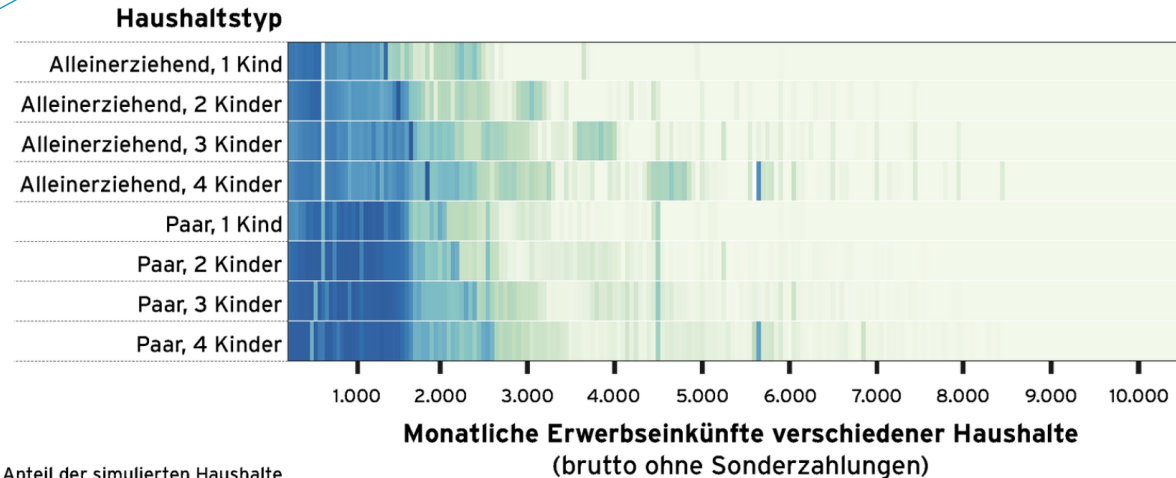
Average Effective Marginal Tax Rate

- For most couples, additional work only pays off from gross €1,600 (1 child), or gross €1,700 (3-4 children).

Wo der Staat alles nimmt

In den blauen Bereichen gibt es besonders viele Haushaltskonstellationen, bei denen der Grenzsteuersatz* mehr als 100 Prozent ausmacht.

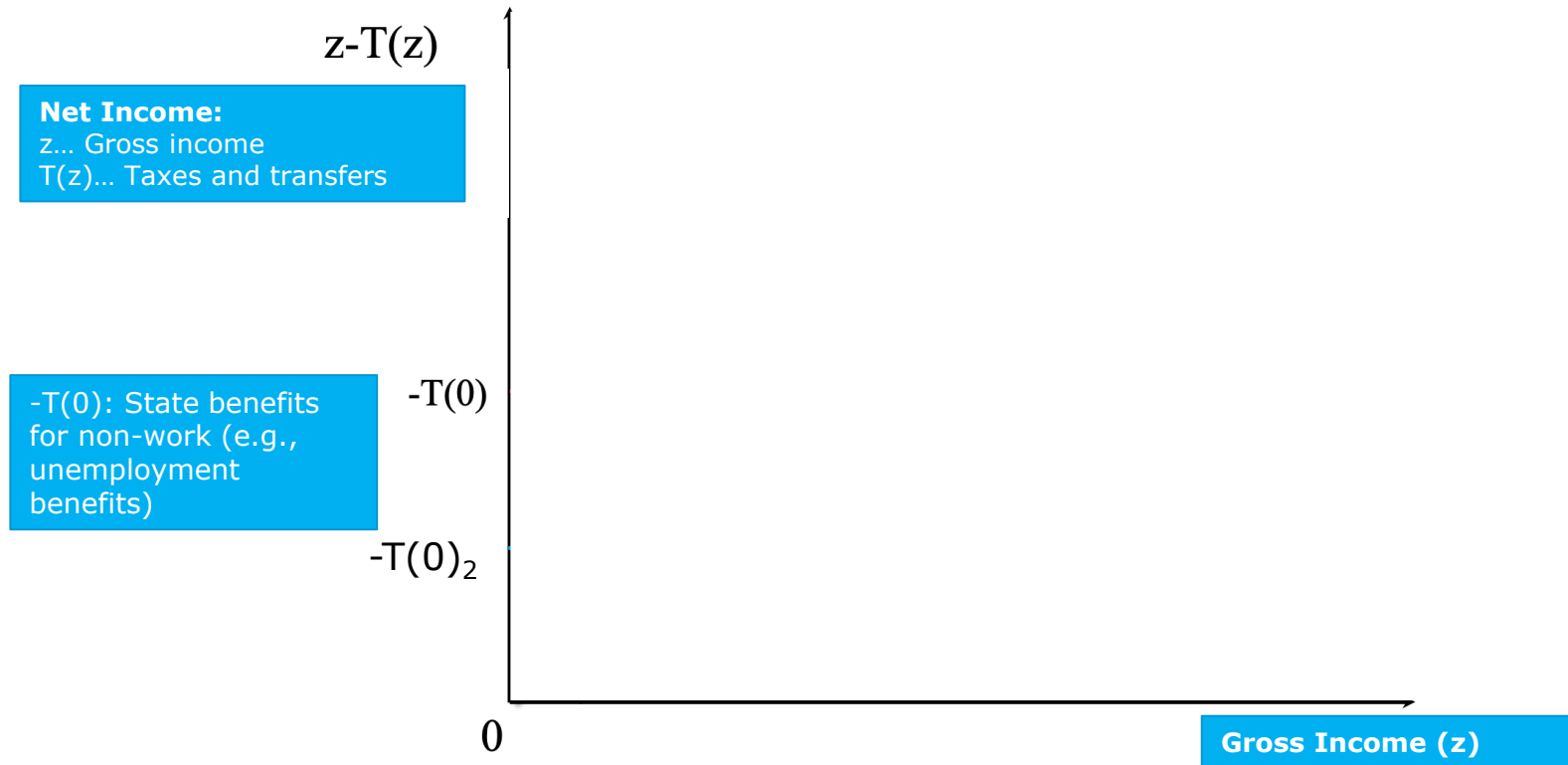
**Effective
Marginal Tax
Rate!!!**



*Der Grenzsteuersatz gibt an, wie stark der letzte Euro des zu versteuernden Einkommens belastet wird.

Simple Gross/Net Model

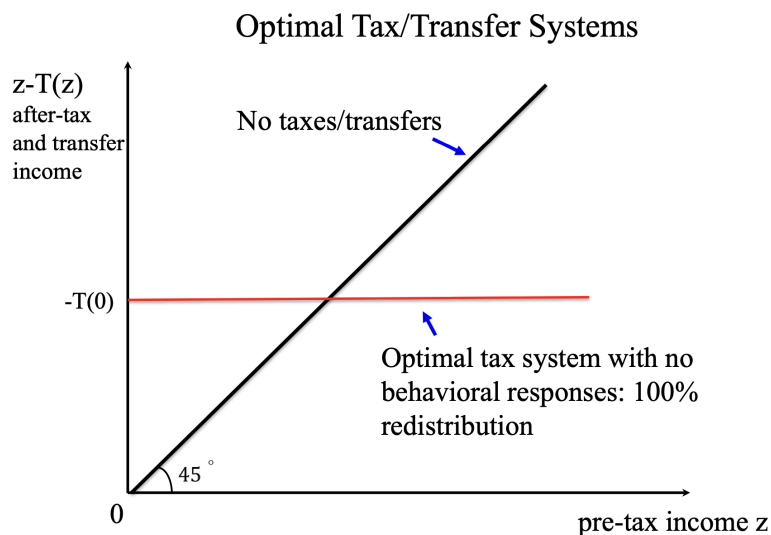
- What does the optimal tax system look like **with and without tax** and with **behavioral responses**?



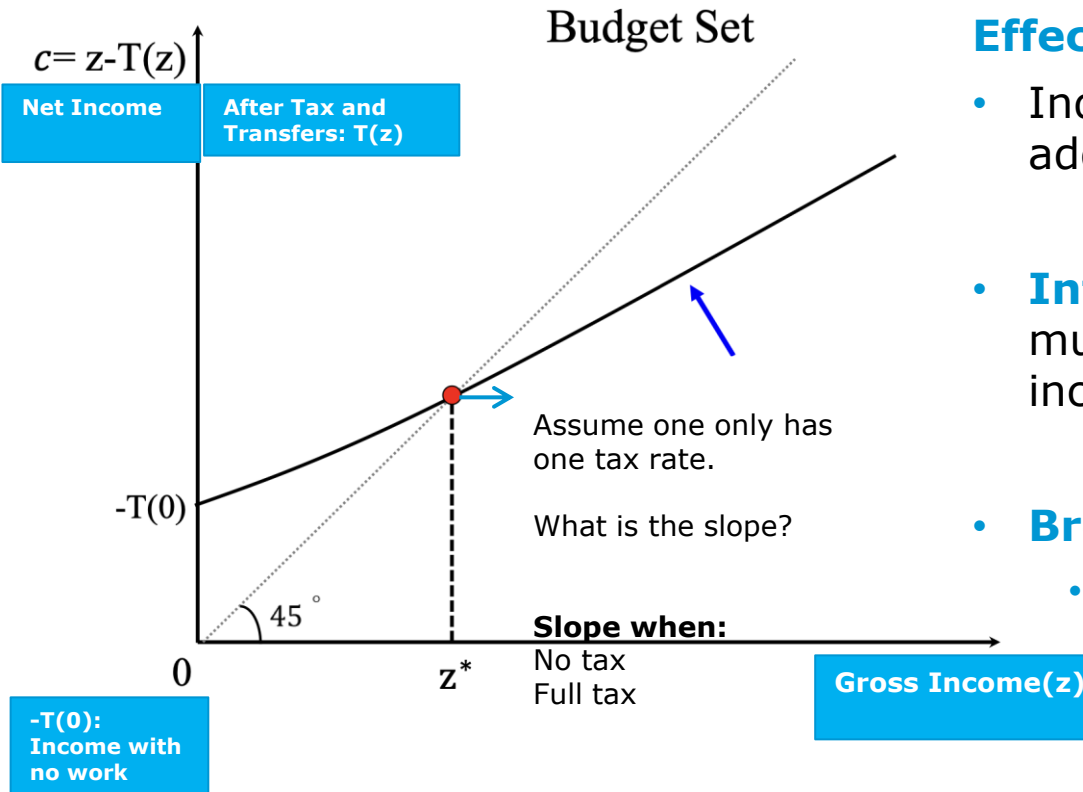
A Few Definitions

What is $T(z)$?

- All taxes and transfers that are paid/received.
- Deducted from gross income!
- Was ist $T'(z)$?
 - Effektive Grenzbelastung!
- Was ist $-T(0)$?
- Staatliche Leistungen bei Nicht-Arbeit (z.B. AL-Geld)



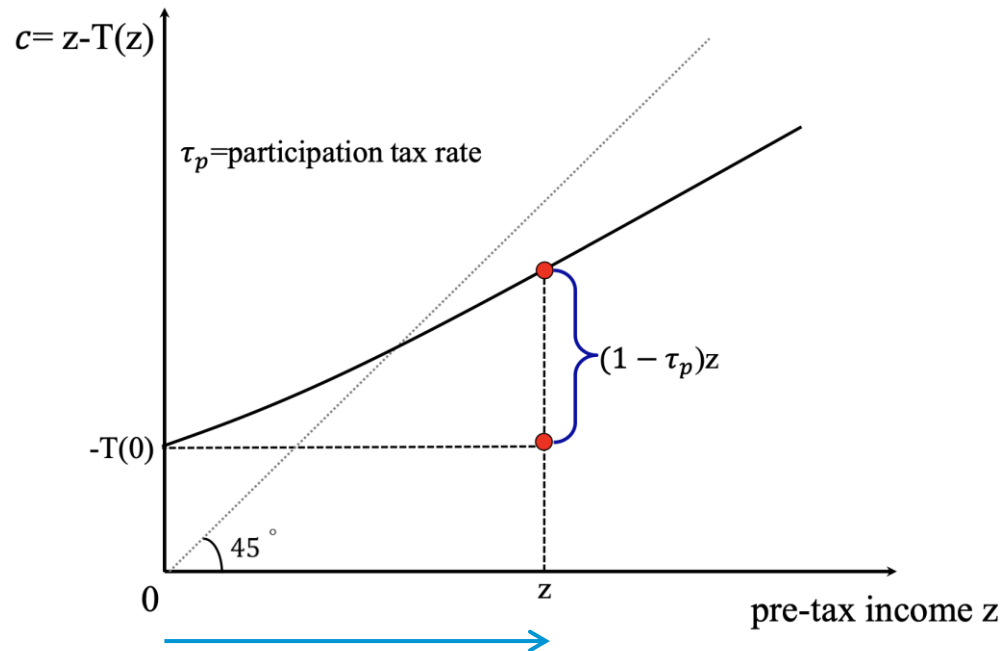
Arbeitsanreiz mit Arbeit



Effective Marginal Burden $T'(z)$:

- Individuals keep $1 - T'(z)$ for every additional EUR 1 of income.
- **Intensive Margin:** Influences how much work should be done (analysis of income tax reactions).
- **Break-Even point: $T(z^*) = 0$**
 - Above the 45-degree line: Incentives to work (v.v. = vice versa).

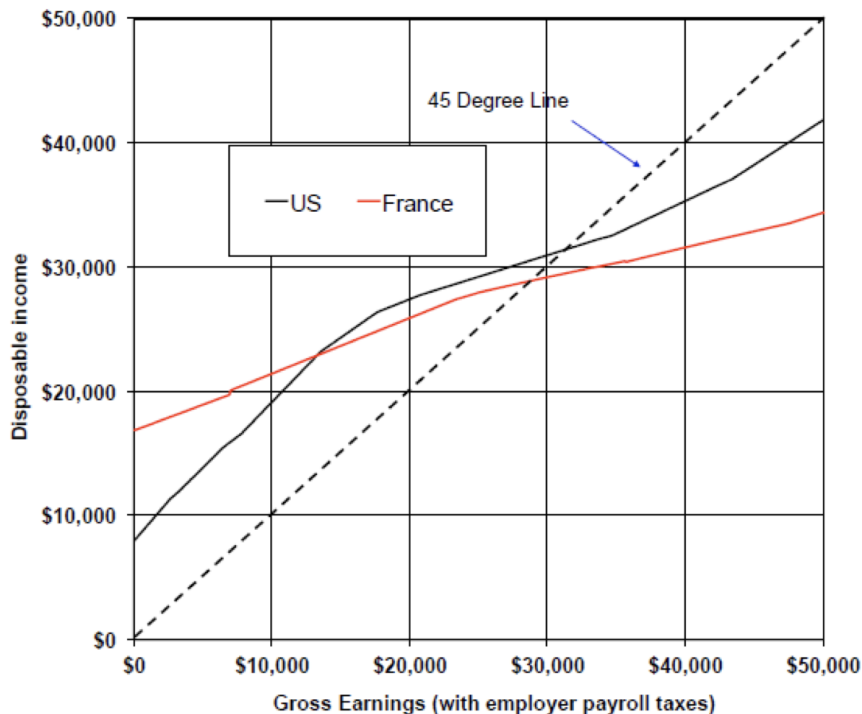
Labor Incentive Without Work



Understood? 2 Examples...

- Individuals keep $(1 - \tau_p)$ of their income when they switch from unemployment to employment (z).
- Participation Tax Burden τ_p (tau):**
- Extensive Margin:** Influences whether or not work should be done.
- $\tau_p = (T(z) - T(0)) / z$
- Net: $z - T(z) = -T(0) + (1 - \tau_p) * z$
- Especially in the **low-wage sector**, the decision of whether to accept a job or not is of greater relevance due to the **transfer withdrawal** and the resulting income tax to be paid.

"Tax-Rate Benefit Cliff & Participation Rate"

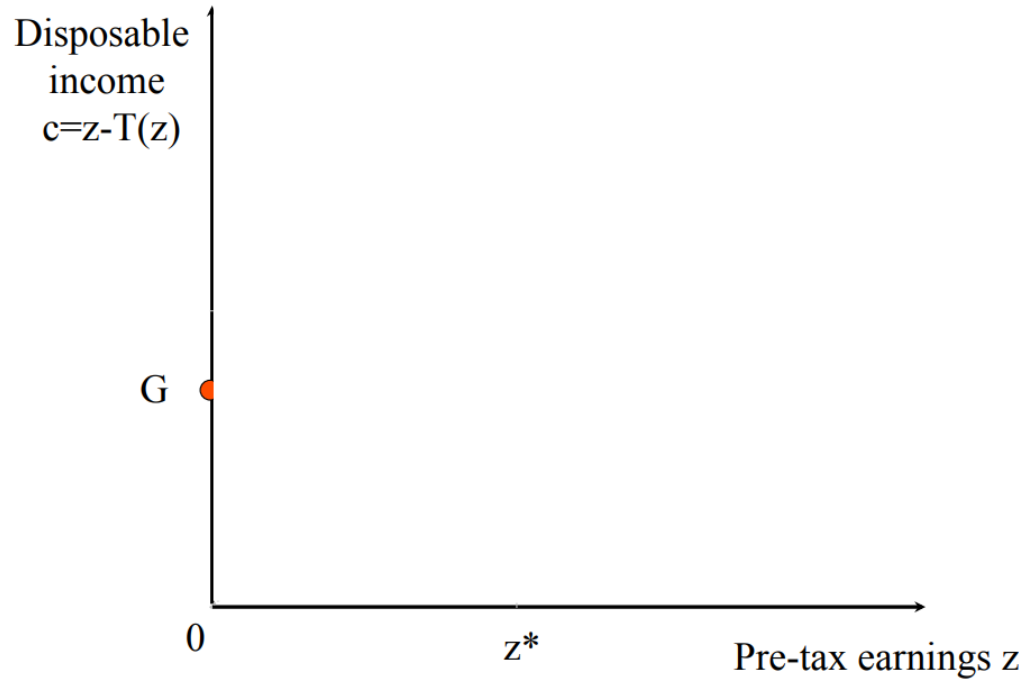


- Who has a higher participation burden?
- Benefit: $1 - \tau_p$:
 - $US > FR$
 - $\Rightarrow \tau_p \text{ FR} > \tau_p \text{ US}$
- Who has a higher marginal burden? $1 - T'(z)$:
 - $US > FR$
 - $\Rightarrow T'(z)$:
 - $FR > US$

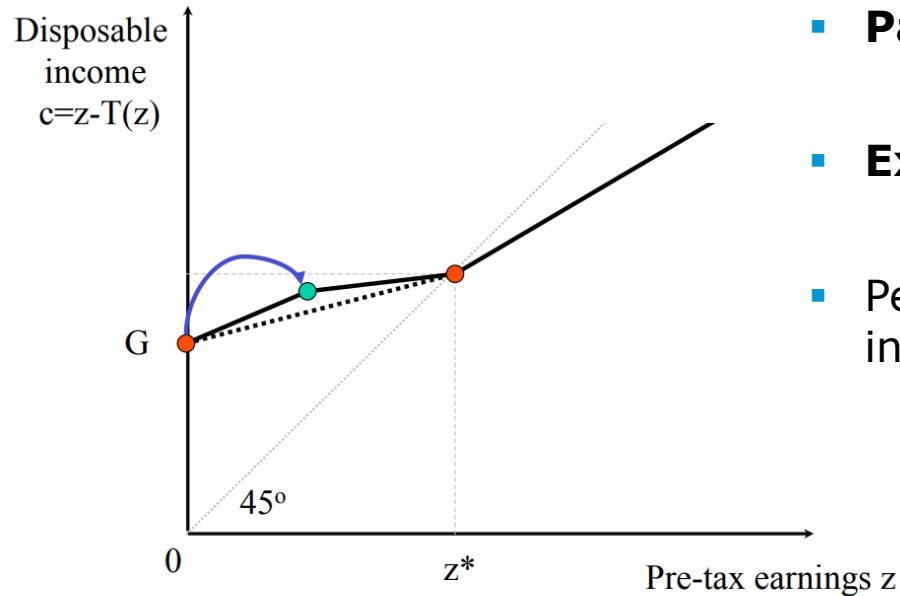
Query riuscita

Optimal Transfers: Wage Subsidies

- Wage subsidies (tax credits) increase

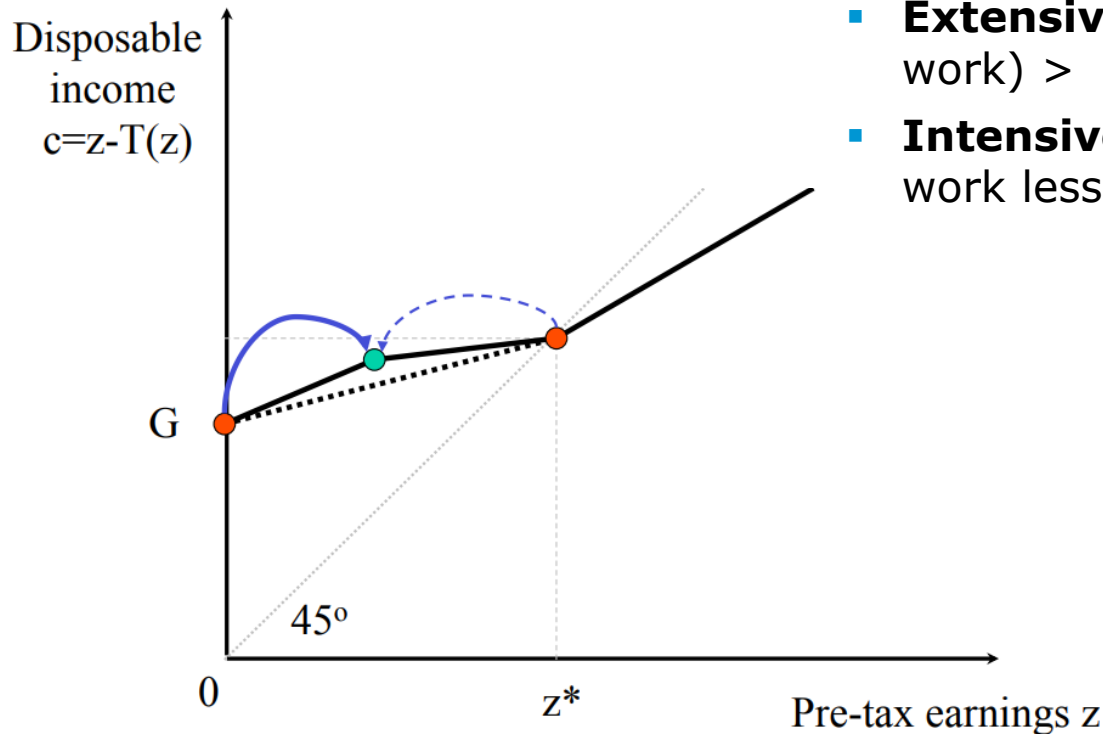


Optimal Transfers: Wage Subsidies



- **Participation Reaction:**
- **Extensive Margin:**
- People now work because there is more income with work.

Optimal Transfers: Wage Subsidies

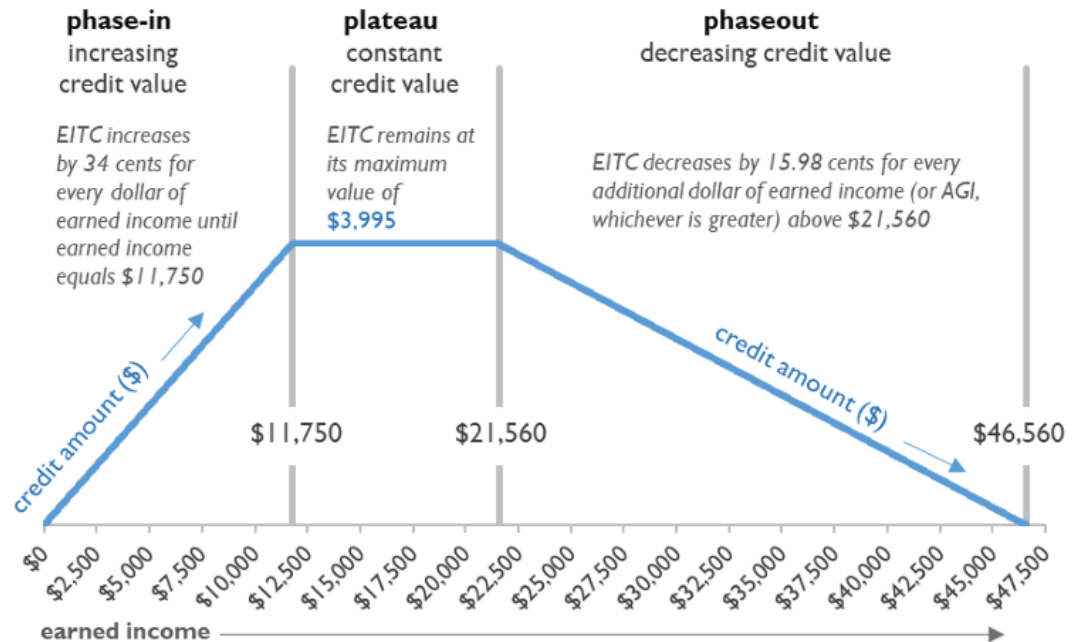


- **Higher Tax Revenue, when**
- **Extensive Margin** (whether one should work) >
- **Intensive Margin** (whether one should work less)---

What exactly does this look like with EITC?

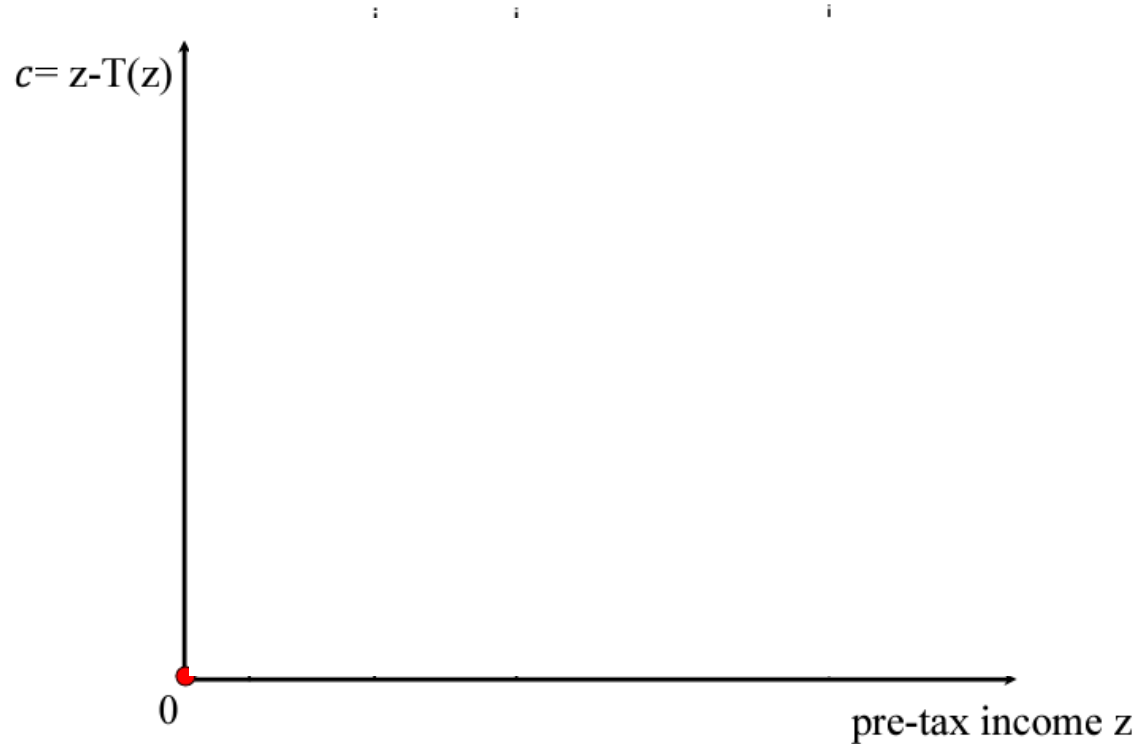
- **Phase-in:**
 - SE: More work, because more work => more EITC
 - IE: Less work
 - **Net: Unclear**
- **Plateau:**
 - Only IE
 - **Net: Less work**
- **Phase Out:**
 - **SE:** Less work, because more work => less EITC
 - **IE:** Less work
 - **Net: Less work**

Figure 2. EITC for an Unmarried Taxpayer with One Child by Income, 2023



What exactly does this look like with EITC?

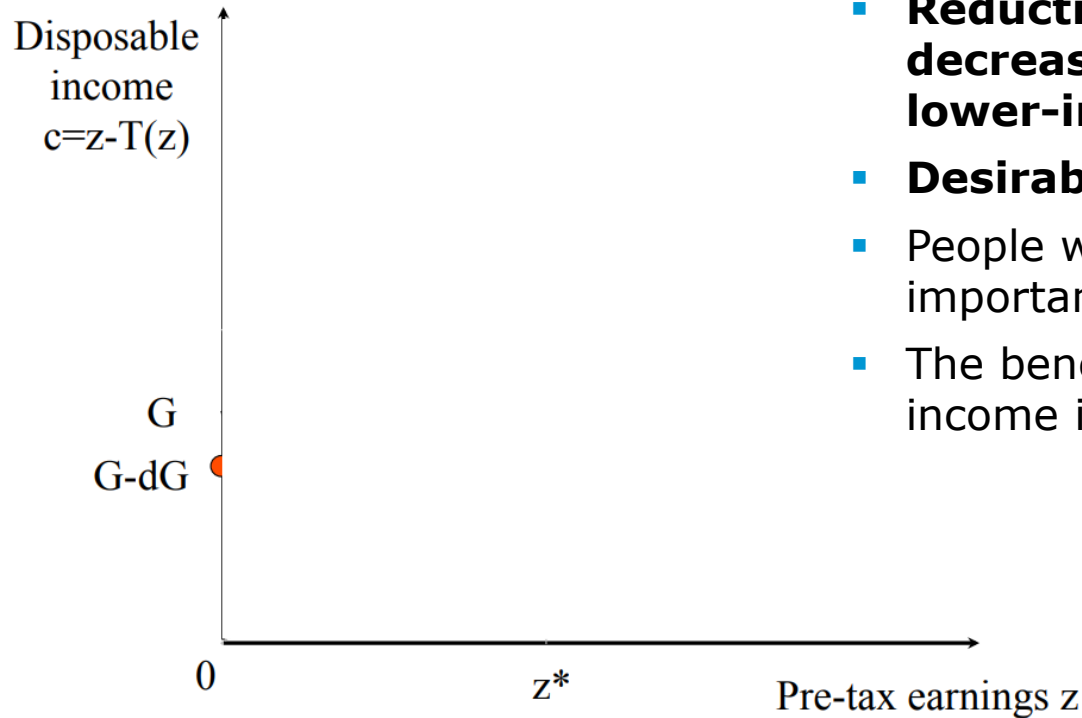
- 3 phases:



Optimal Transfers: The "Laziness" Argument

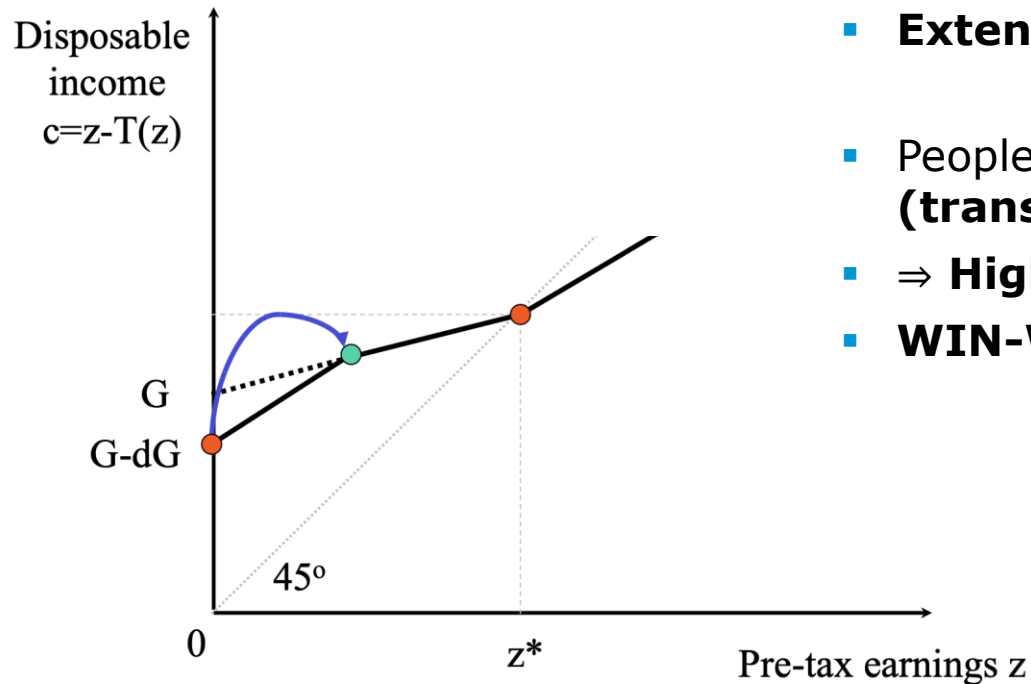


Optimal Transfers: The "Laziness" Argument



- **Reduction of transfer payments and decrease of marginal tax rates for lower-income groups**
- **Desirable, when:**
- People with no income are of lesser importance
- The benefit of EUR 1 for people with no income is $<$ EUR 1 for everyone

Optimal Transfers: The "Laziness" Argument



- **Participation Reaction:**
- **Extensive Margin**
- People **start working** because of **lower (transfer) income**.
- $\Rightarrow$ **Higher government revenues**
- **WIN-WIN?**

Optimal Transfer Payments

- If **people without income** are viewed as **less needy** (than the average), then:
- **Low basic income with rapid reduction of benefits is optimal.**
- If **low-income earners** are viewed as **more needy** (than the average) and the **labor supply reaction is stronger** on the **extensive margin**, then:
 - **Higher wage subsidies and a slow reduction of benefits are optimal.**
 - **—**

Which option do you prefer?

- **Option 1:** Means-tested transfer of €1,000 per month for those without income. Benefit reduced by €0.20 per additional earned Euro. Financed by progressive tax: 20% on income above €60,000.
- **Option 2:** Universal basic income of €1,000 per month for everyone. Financed by flat 20% tax on all income.

Net Income Comparison

Option 1: Transferleistung mit progressiver Steuer

Person	Income	Transfer from Government	Taxes Paid	Net Income
A	\$0	\$12,000	\$0	\$12,000
B	\$30,000	\$6,000	\$0	\$36,000
C	\$60,000	\$0	\$0	\$60,000
D	\$90,000	\$0	\$6,000	\$84,000

Option 2: Grundeinkommen

Person	Income	Transfer from Government	Taxes Paid	Net Income
A	\$0	\$12,000	\$0	\$12,000
B	\$30,000	\$12,000	\$6,000	\$36,000
C	\$60,000	\$12,000	\$12,000	\$60,000
D	\$90,000	\$12,000	\$18,000	\$84,000

- => Consider taxes and transfers together!

Schlussfolgerung: Greg Mankiw

- Many people tend to prefer **Option A**
- Options 1 and 2 are **identical**, only framed differently!
- Plan B is **administratively simpler**.
- Always consider **taxes and transfers together!**.
- **Plan A:**
 - Lower taxes, progressive taxes, progressive transfers.
 - But equivalence is only clear when taxes and transfers are considered together.
 - Media often tend to consider taxes and transfers in isolation!
- Example: <https://www.cpic-global.net/country/AUT>

Basic Income Example



Empirical Evidence

Basic income experiments I

- **How does labor supply react when basic income is introduced?**
- Extreme form of **negative income tax**.

Everywhere basic income has been tried



Source: Vox Research

Vox

Basic Income Experiments II

- **1970s US NIT Experiments** (small scale):
 - Men: labor supply ↓ 7%
 - Women: labor supply ↓ 17%
- **Newer Experiments** Small/No Effects on Labor Supply (Alaska, Finland, Barcelona)
 - Most effects are only on the intensive margin (reduction in hours), but not on leaving the labor market.
 - Substitution to other "occupations" (childcare, education, volunteer work)
- **Much Evidence for strong Effects on Well-being and Health**
- Overview: <https://basicincome.stanford.edu/research/ubi-visualization/>
 - [pre analysis plan mein grundeinkommen.pdf](#)

Unemployment Insurance Example



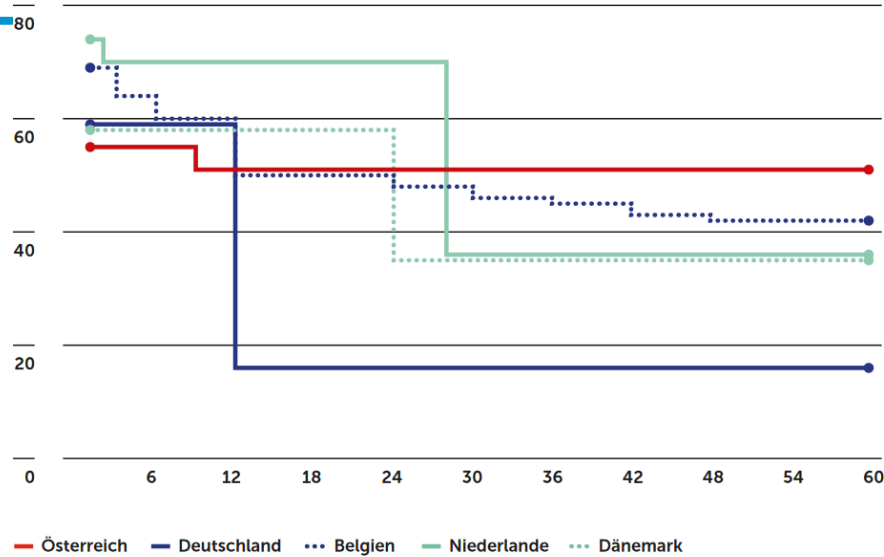
Empirical Evidence

Unemployment Insurance Example

- Insurance against periods of unemployment
- **Important on an individual level:**
- Buffering of income shocks, poverty reduction
- Allows for longer job search → better job-matches
- **Possible problematic incentives:**
- Lengthens phases of unemployment
- Not enough incentive to work at all

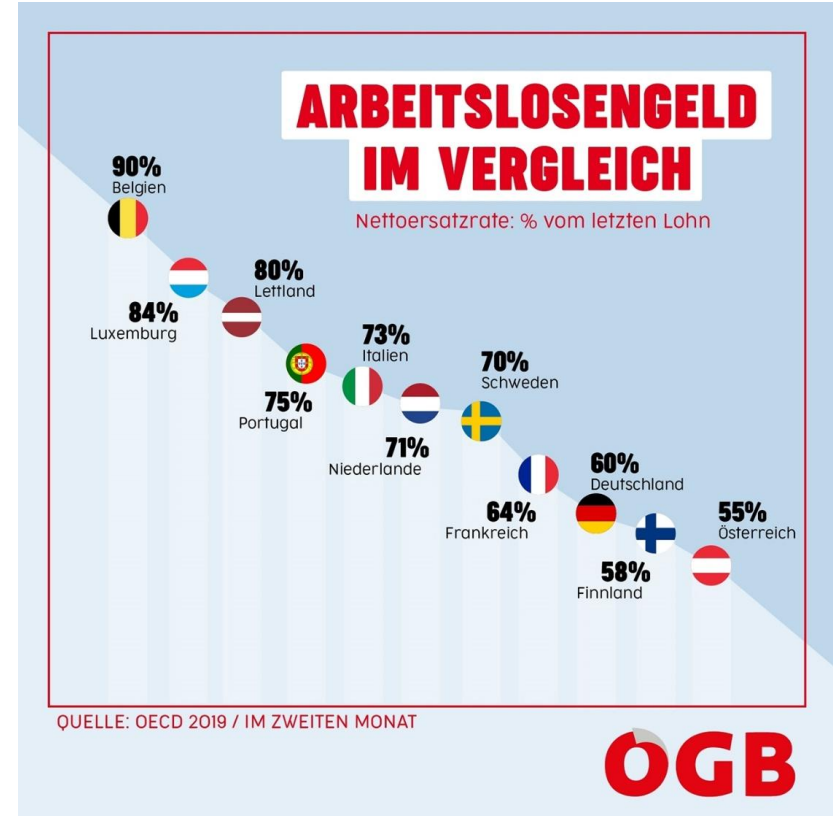
Arbeitslosenunterstützung nach Bezugsdauer

– in Prozent des Letzteinkommens nach Steuern, Bezugsdauer in Monaten



Quellen: Eigene Berechnungen, OECD.

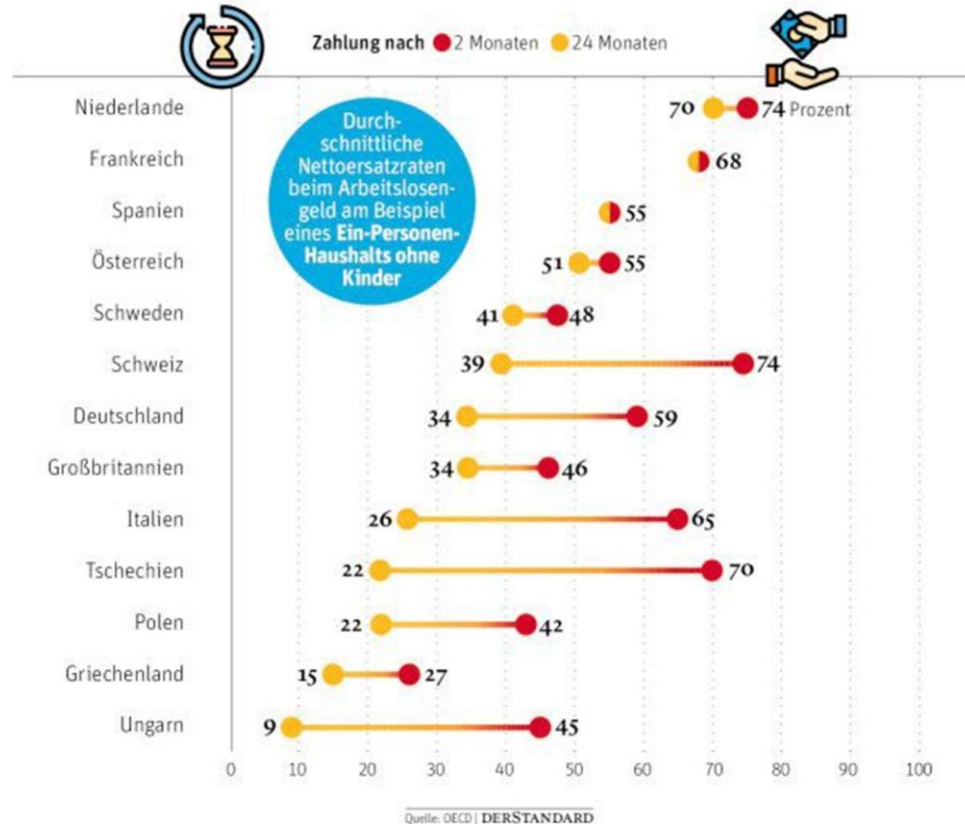
Anmerkung: Arbeitslosenunterstützung (Arbeitslosengeld, Notstandshilfe und andere) ohne Berücksichtigung der Wohnbeihilfe für einen Durchschnittsverdiener, Single ohne Kind.



OGB

Leistung für Arbeitslose im Vergleich

2019, in Prozent des Nettoeinkommens



- Benefits in Austria are rather low
- However, they do not decrease significantly over time (due to emergency assistance)

Empirical Evidence I

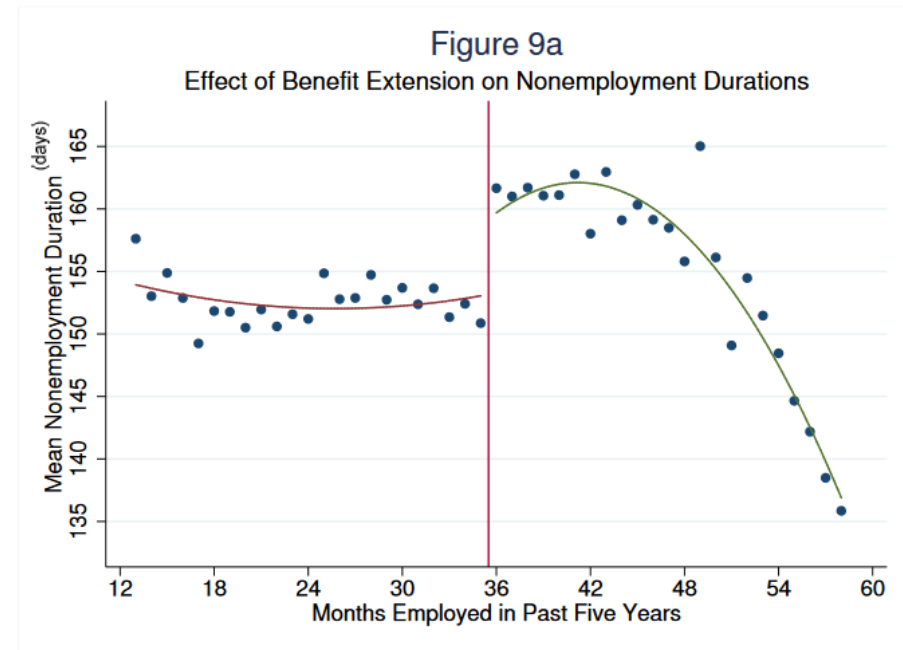
Card, Chetty, Weber (2007) „Cash-on-Hand and Competing Models of Intertemporal Behavior: New Evidence from the Labor Market”

Setup

- Research Question: longer eligibility period → longer unemployment??
- Jump in the eligibility period for unemployment benefits in Austria
- Those who were employed for more than 156 weeks in the last 5 years are eligible to receive unemployment benefits for 30 weeks instead of 20 weeks

Results

- Relatively small but clearly positive effect on the duration of benefits: approx. 7 days



Empirical Evidence II

Arash, Nekoei; Andrea, Weber (2017) *Does Extending Unemployment Benefits Improve Job Quality?*

Setup

- Research Question: Does a longer eligibility period lead to better job matches?
- Unemployed over 40: 39 instead of 30 weeks

Results

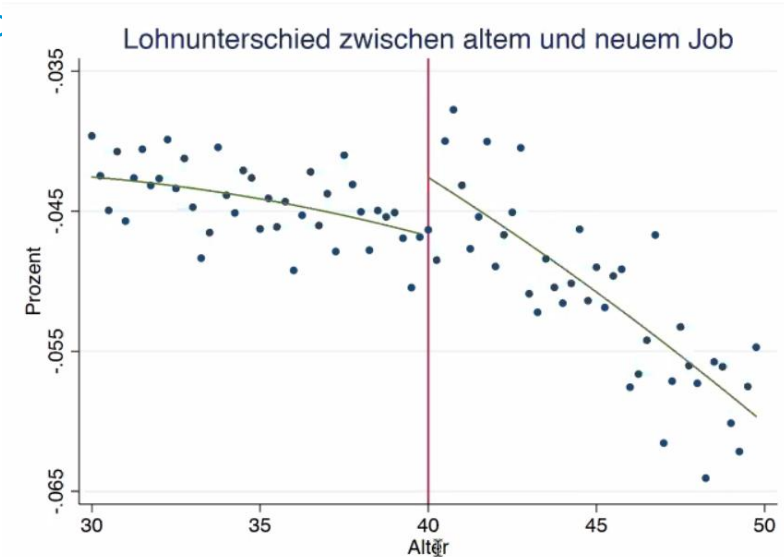
Unemployment is extended

BUT better jobs (higher wages)

In other studies, the positive effect is not clear, as there are two opposing mechanisms:

- More time to look for a better job)
- Longer unemployment reduces job quality)

[VOX-artikel Zusammenfassung](#)



Special Case: Information Asymmetries



Empirical Evidence

- Bis jetzt sind wir davon ausgegangen, dass Individuen vollkommene Informationen über den gesamten Arbeitsmarkt besitzen.
- **Aber was wenn nicht?**

Subjective vs. Objective Information

- **Current research** addresses important questions about inequality and views on fairness by combining **subjective perceptions with objective facts**(Almås et al. 2017, Karadja et al. 2017, Epper et al. 2020,...).
- Relevant for the optimal redistribution measure!

„Biased Wage Expectations and Female Labor Supply“ (Blesch et al., 2023)



- SOEP: Deutsches Sozio-ökonomisches Panel (SOEP)
- Full-time Wage Expectation:

Example: Question 1 to a woman that is currently employed full-time

Current working hours:

Suppose you continue to work full-time in the coming years, regardless of whether you are actually planning a work reduction or anything similar. Please think about full-time jobs that you can perform with your qualification. If, in reality, you are planning to reduce your workload, please still assume for the moment that you continue to work full-time in the next years.

What do you think is your gross monthly income ...

- in one year?
- in 2 years?
- in 10 years?

Part-time Wage Expectation

Example: Question 2, counterfactual question to a woman that is currently employed full-time

Hypothetical working hours:

Please imagine you were to switch to a part-time job from now on, working 20 hours per week. Please only consider part-time jobs that you could carry out with your current level of qualification.

- What gross monthly income do you expect to earn when working part-time at 20 hours per week? Probabilistic add-on (II)

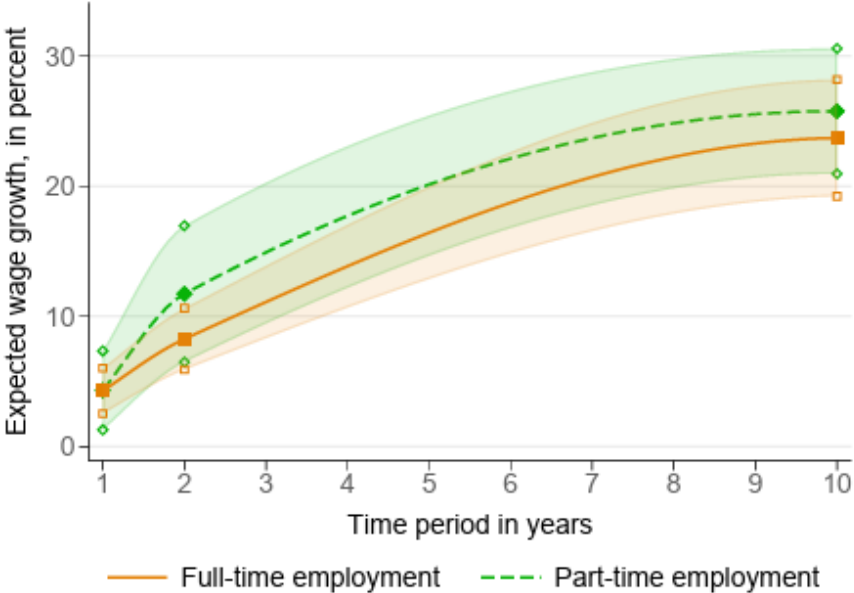
Now suppose that you continue to work part-time in the coming years, working 20 hours per week.

What do you think is your gross monthly income ...

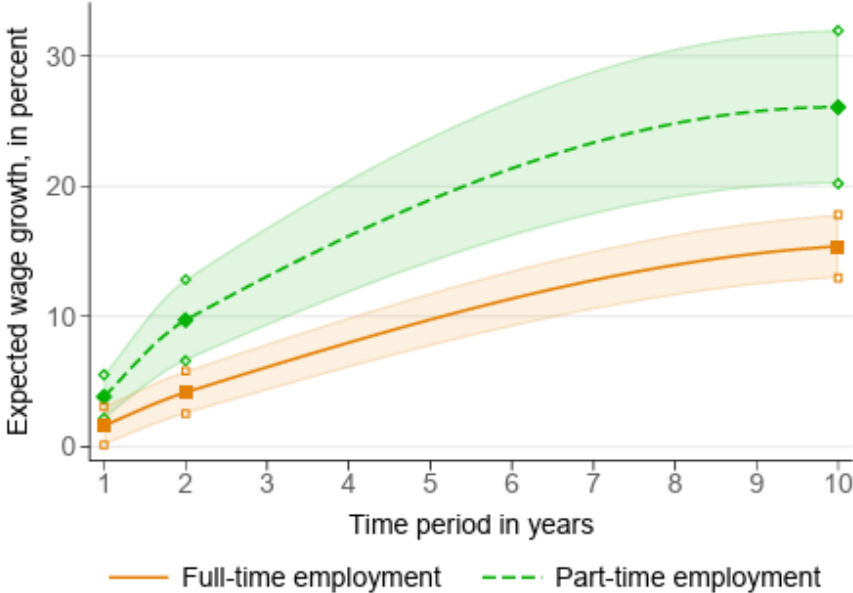
- in one year?
- in 2 years?
- in 10 years?

Expectations for Full-time/Part-time Wage Development

(a) Full-Time Workers



(b) Part-Time Workers



- What do you think the **actual wage developments** look like between **part-time and full-time**?

Only for full-time!

■ Actual Returns:

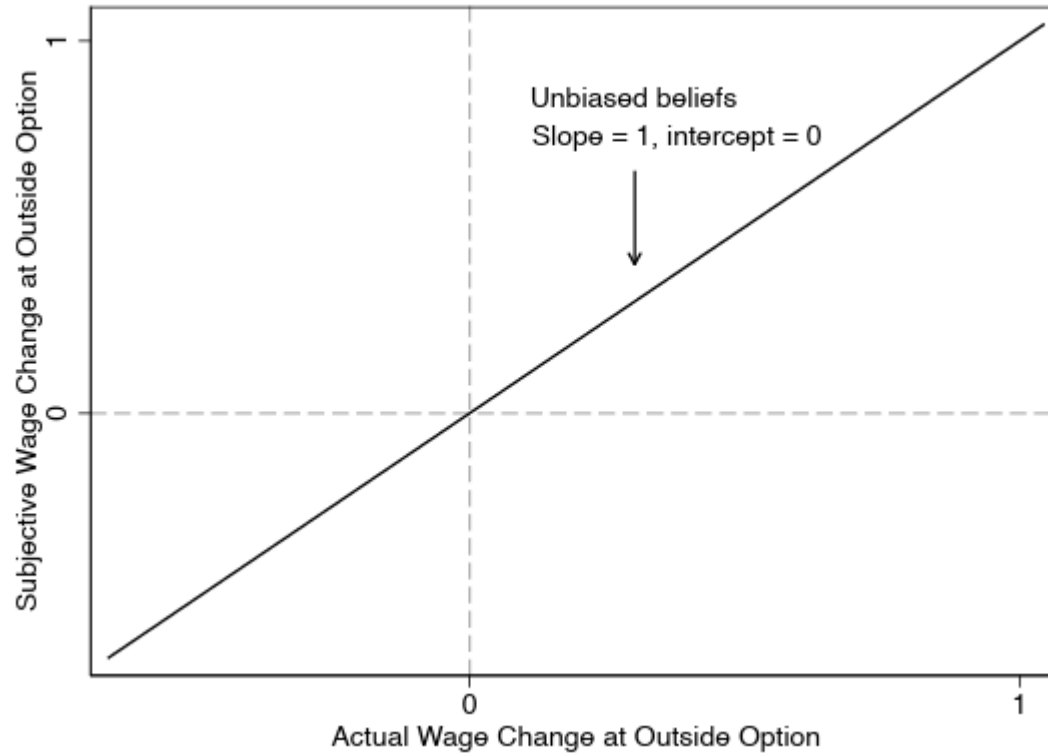
	Low Education	Medium Education	High Education
Experience in Part-Time	0.003	-0.001	0.002
Experience in Full-Time	0.009***	0.014***	0.014***
N	15449	31025	11256

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ SOEP v35. 1992-2017. FE and Control Functions based on the four instruments defined above. Model accounts for the current employment state and includes an indicator for living in East Germany.

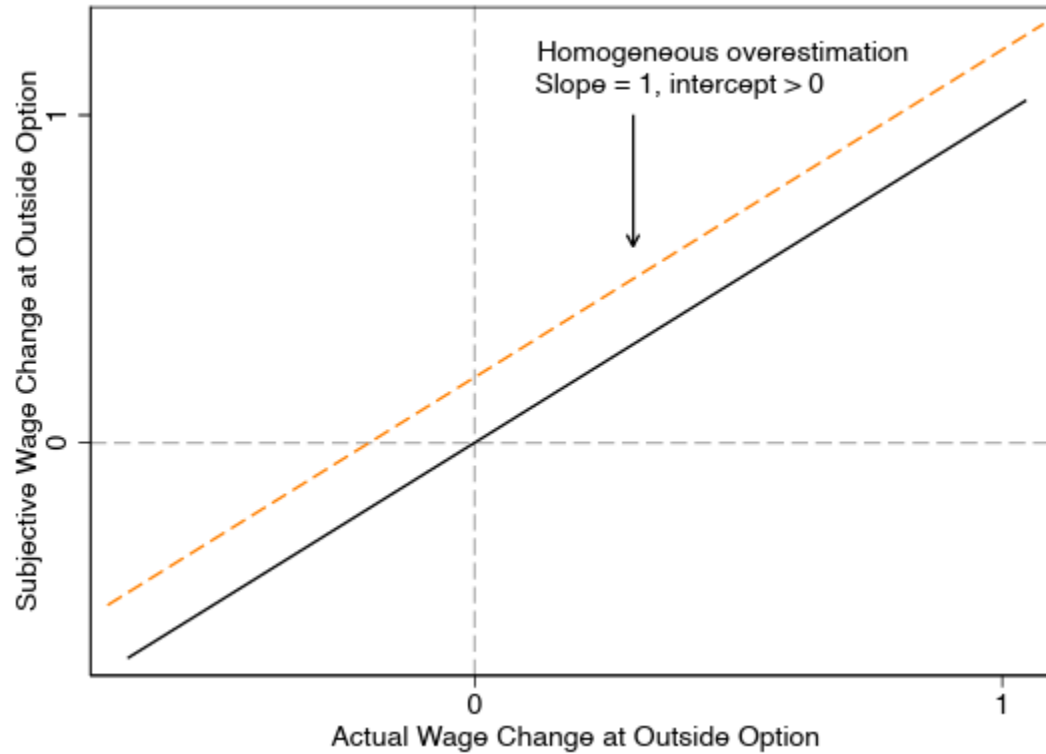
Worker beliefs about outside options (Jäger et al., 2024)

- Standard labor market models assume that employees **have accurate information about wage distribution**
- „Outside options" are known

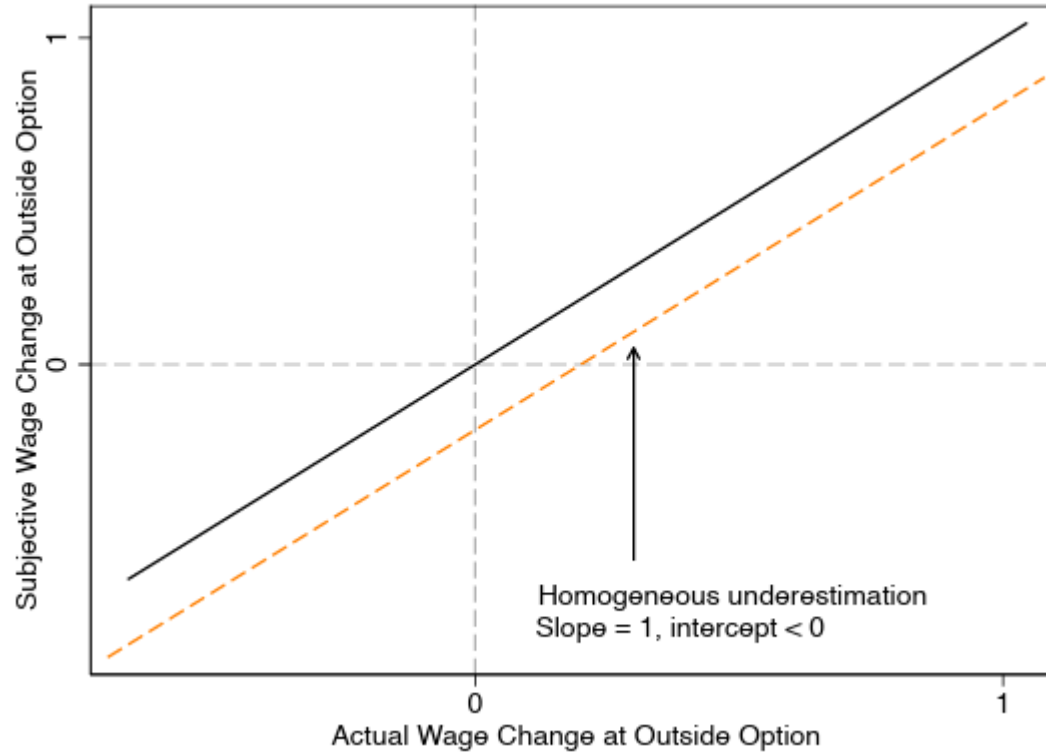
Subjective = Objective



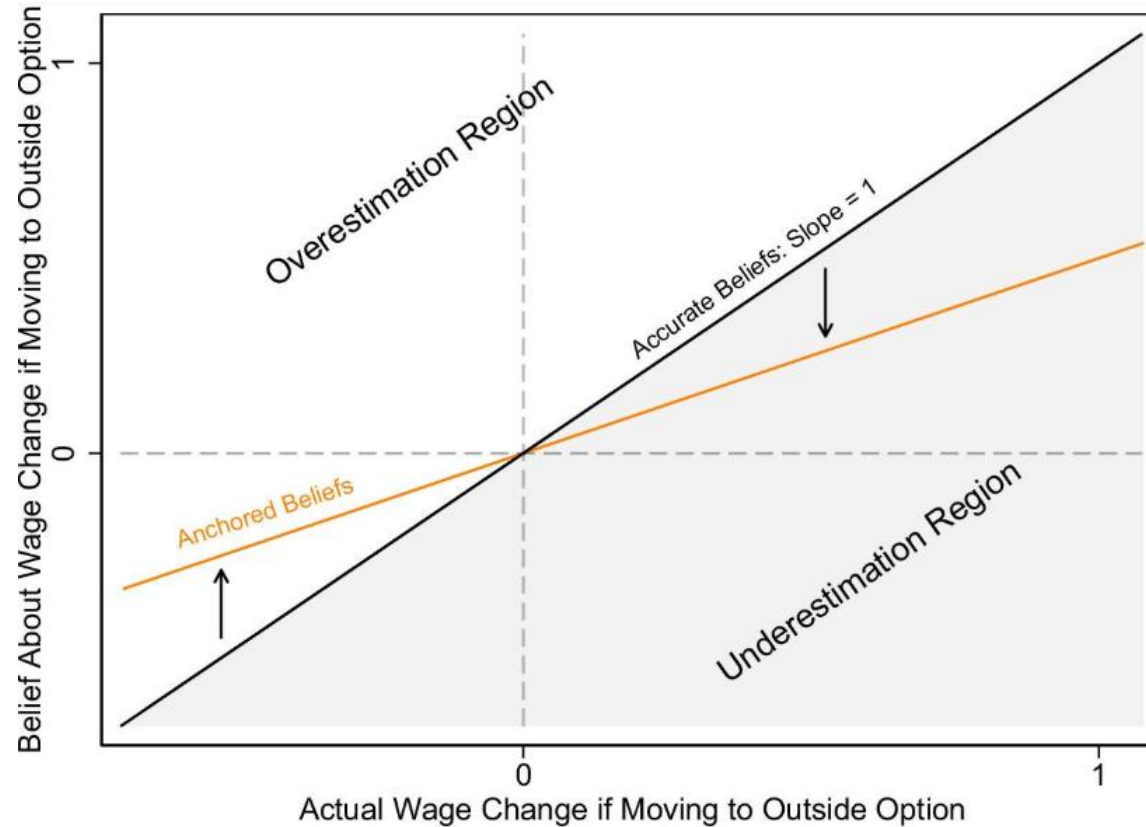
Overestimation



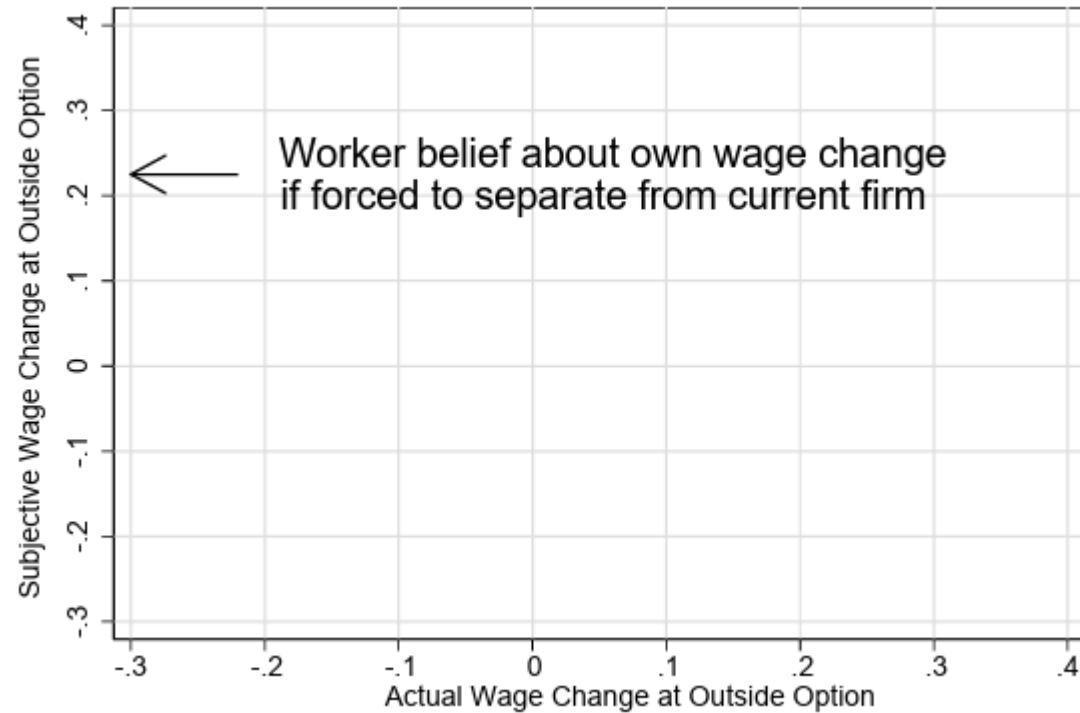
Underestimation



„Anchoring“



Subjective Assessment?

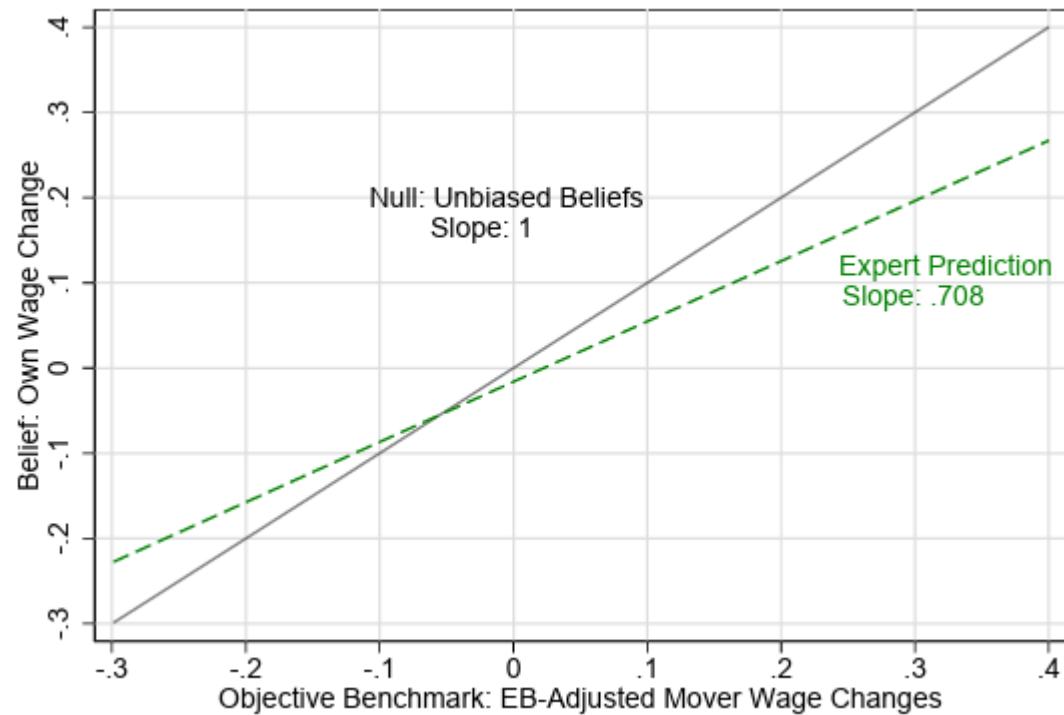


Beliefs about Outside Options

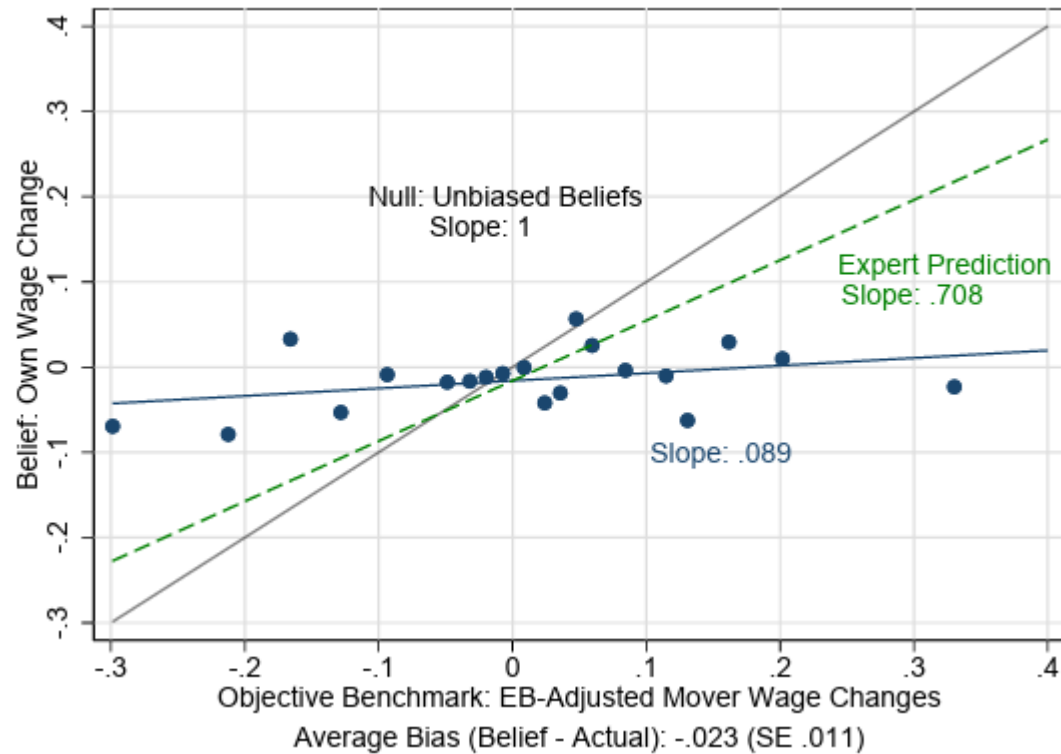
- „ **Imagine you had to quit your current job** and had three months to find a job with a **different employer in the same profession.**
- Do you believe that you would earn **on average more or less gross per month than in your current job?**
 - + how much more/less

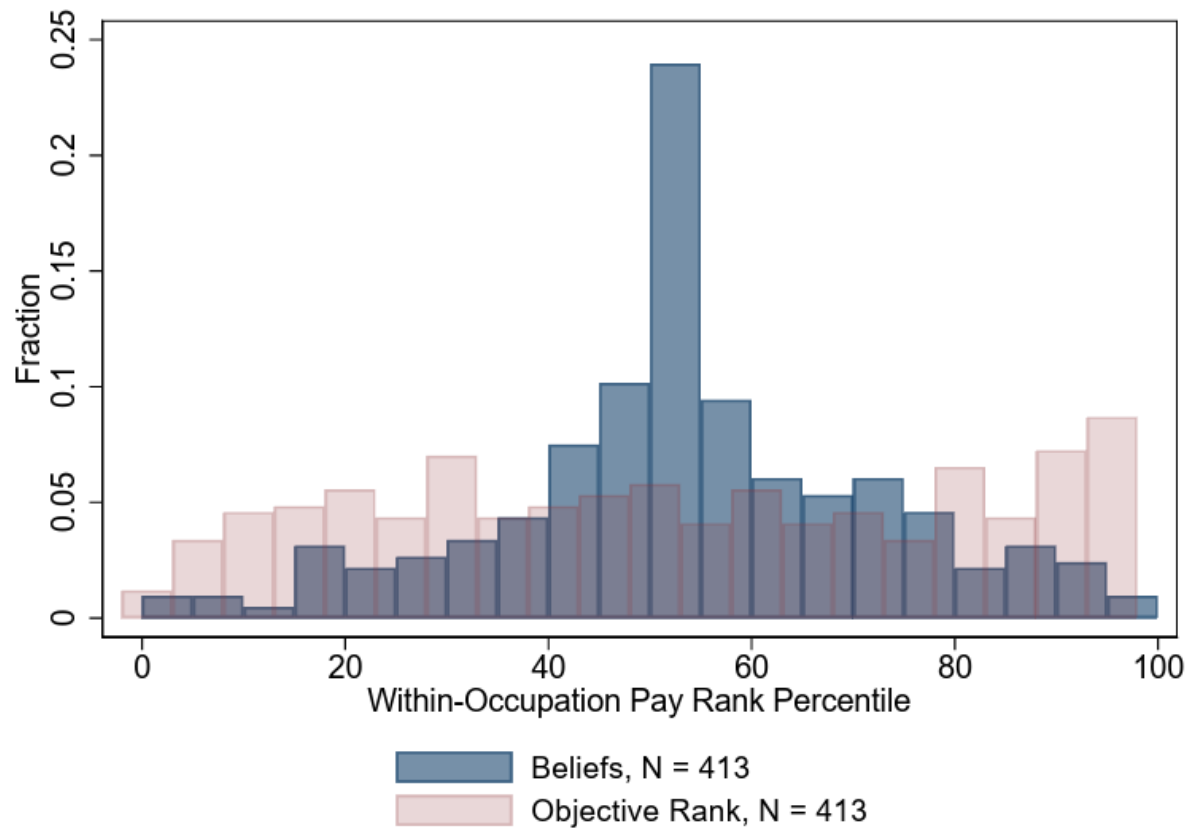
- **Repräsentative** Stichprobe der deutschen Bevölkerung
 - Hohe Qualität: Persönliche oder computergestützte Telefoninterviews
 - 1.604 Befragte, mit Panelstruktur über beide Wellen
- **Stichprobenmerkmale**
 - Verknüpfung mit administrativen Arbeitgeber-Arbeitnehmer-Daten (SOEP-ADIAB)
- **Zusätzliche Erhebungen**
 - Befragung akademischer Experten
 - Zusätzliche Umfrage zur Überprüfung der Robustheit

Expectation Experts

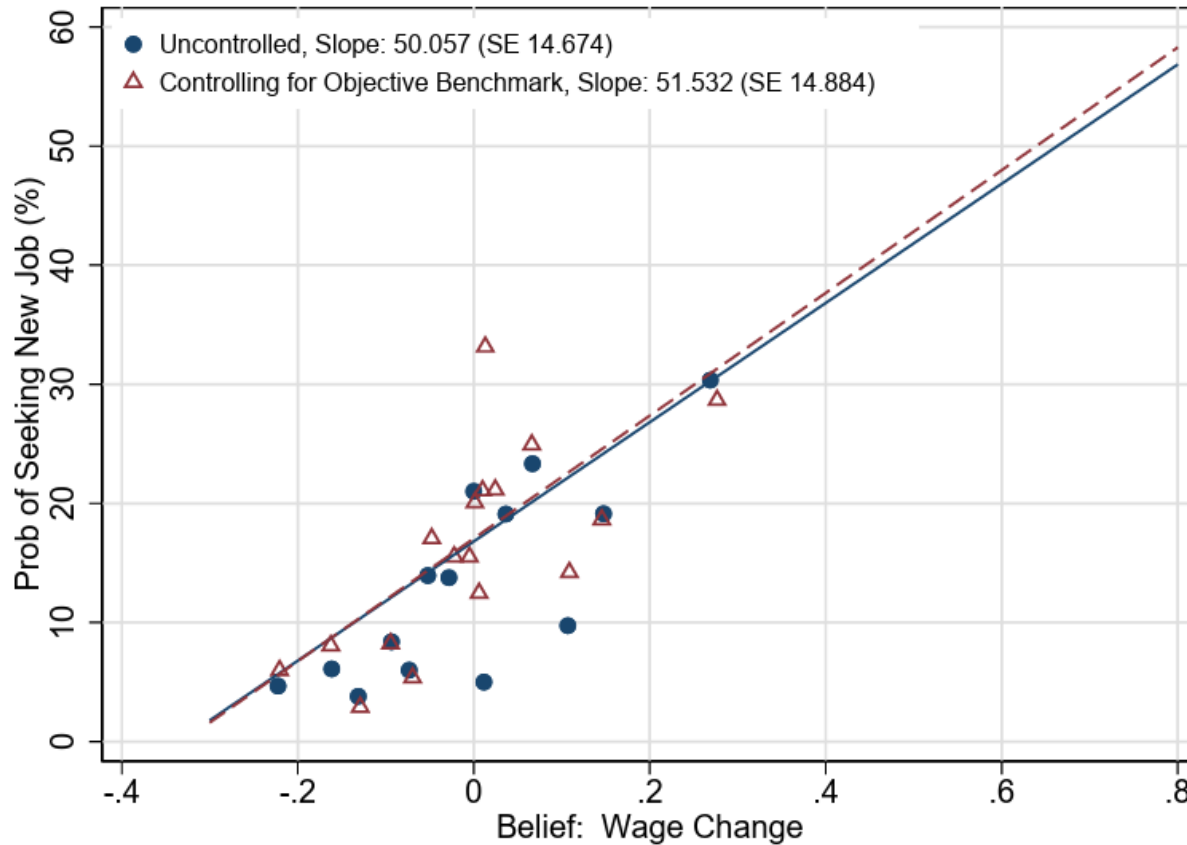


Result



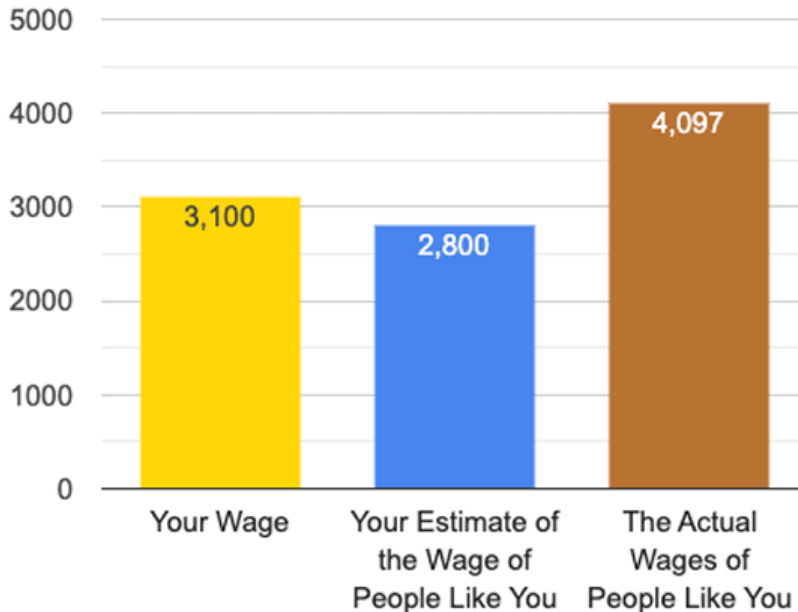


Beliefs & Job Search



Information increases job search

Average Monthly Earnings of People Like You (In Dollars)



	(3) Intended Quit Probability	(4) Intended Search Probability	(5) Intended Negotiation Probability	(6) Intended Neg Magnitude (No Neg = 0)	(7) Reservation Wage Cut
Treated × Pre-Treat Estimation Error	-0.121*** (0.043)	-0.066 (0.044)	-0.187*** (0.048)	-0.029*** (0.005)	0.036 (0.035)
Treated	1.152 (1.096)	2.499** (1.115)	0.945 (1.295)	0.166 (0.132)	-2.718 (2.014)
Pre-Treat Estimation Error	-0.036 (0.032)	-0.056* (0.030)	0.088** (0.034)	0.007* (0.004)	-0.028 (0.033)
Constant	22.823*** (0.776)	24.235*** (0.768)	39.134*** (0.923)	6.942*** (0.092)	11.621*** (1.998)
Mean Dep. Var.	24.06	26.08	39.66	7.08	10.36
Nb. obs	3.206	3.206	3.206	3.206	3.204
IV: Belief: % Wage Change at OO	0.269*** (0.078)	0.223*** (0.079)	0.382*** (0.092)	0.059*** (0.010)	-0.176 (0.167)
Constant	21.739*** (0.725)	24.083*** (0.734)	37.261*** (0.856)	6.659*** (0.090)	11.381*** (1.557)
Control Group Mean	23.055	24.595	38.574	6.895	11.799
First-Stage F-Stat	171.515	171.515	171.515	171.515	170.520

Social Positions and Fairness Views on Inequality

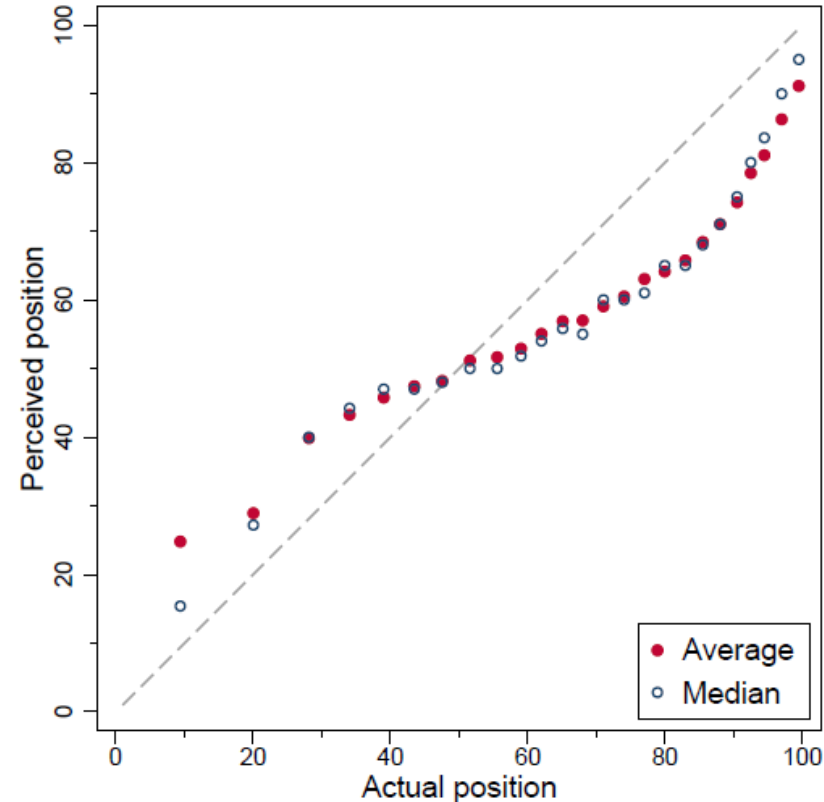
Hvidberg, K. B., Kreiner, C. T., & Stantcheva, S. (2023)

Research Question: How do social positions influence the perception of inequalities and views on the fairness of inequalities?

- Linking **administrative data** and **survey** data in Denmark
- Income differences **within the same education level** or between colleagues are considered the **most unfair**.
- Views on the fairness of inequalities **change** when people are **experimentally informed about their actual social position**.

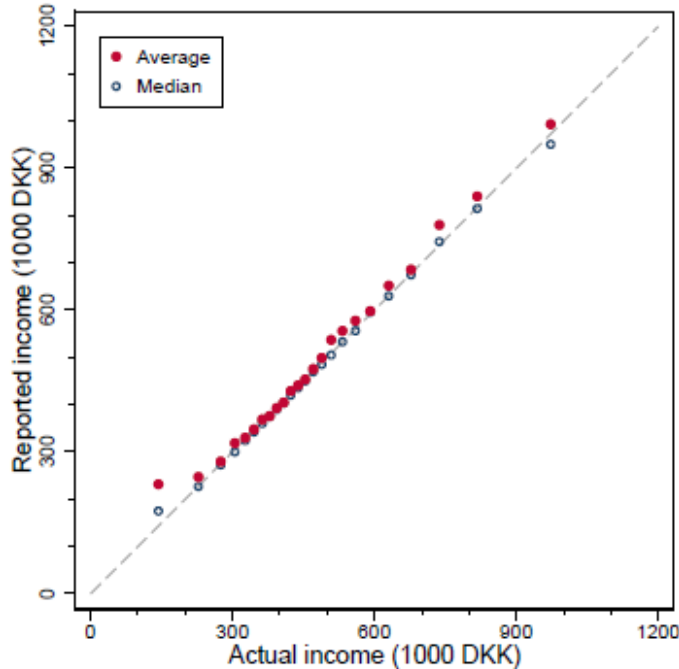
Perceived vs. Actual Position in the Same Birth Year: "Center Bias"

- What is shown here?
- People with incomes below the median overestimate their rank, while people above the median underestimate their rank
- What influences these distorted perceptions?
- **Is it a misperception of one's own income or others' income?**

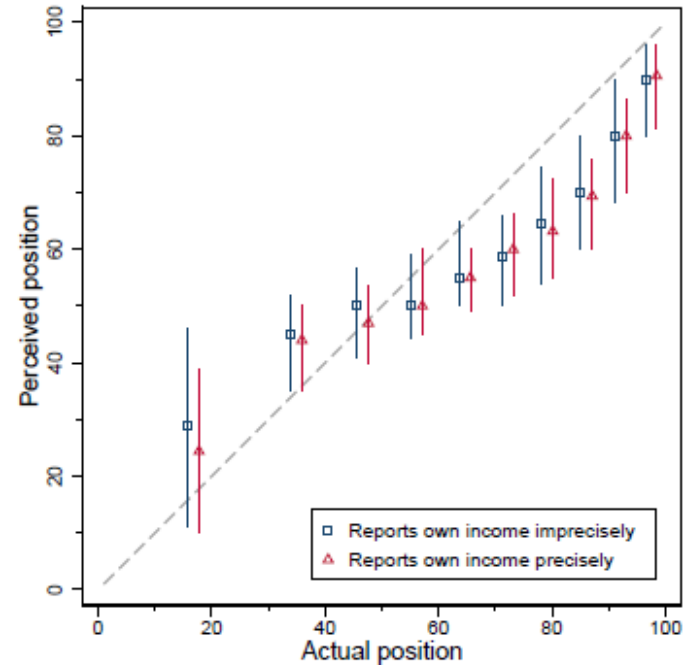


Do false perceptions of one's own income play a role?

(A) ACTUAL VS. REPORTED INCOME



(B) PERCEIVED VS. ACTUAL POSITION

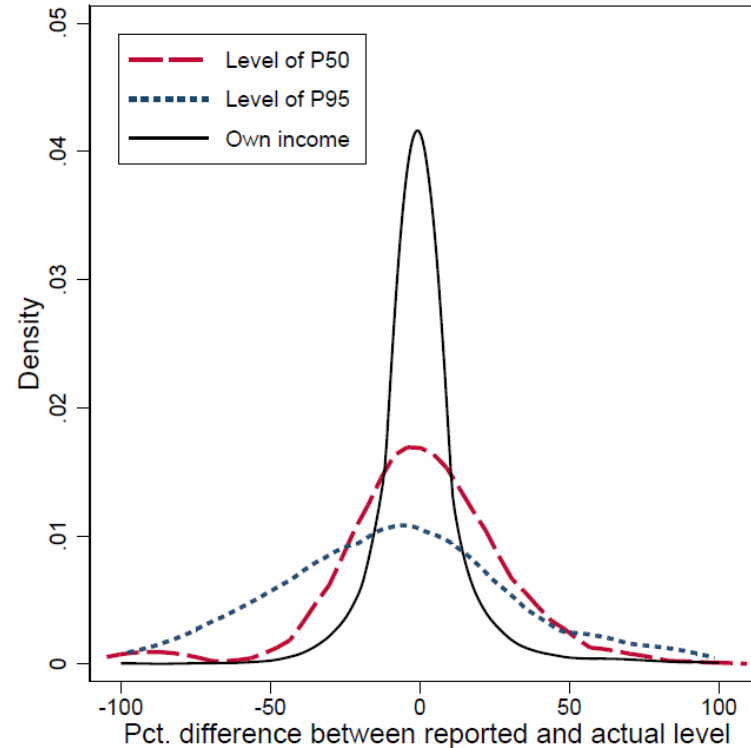


■ ... No!

False Perception of Others!

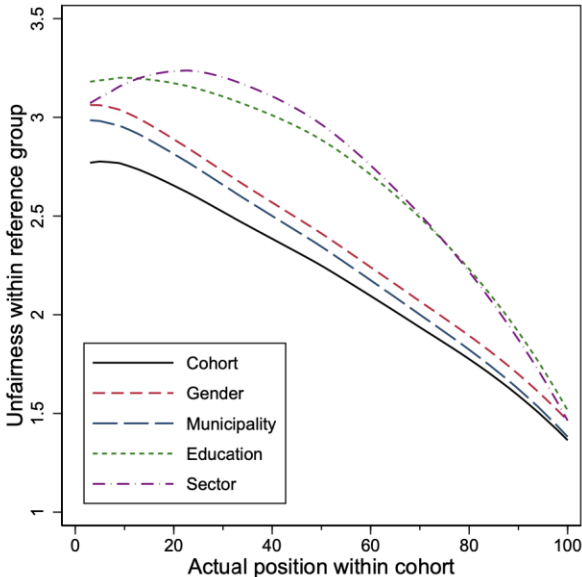
(A) MISPERCEPTION SIZE

- Median incomes are perceived better than extreme incomes.
- Why?

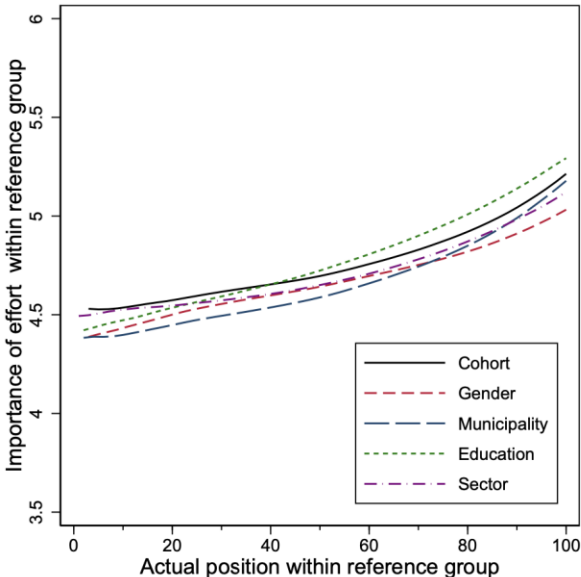


Fairness views dependent on ...

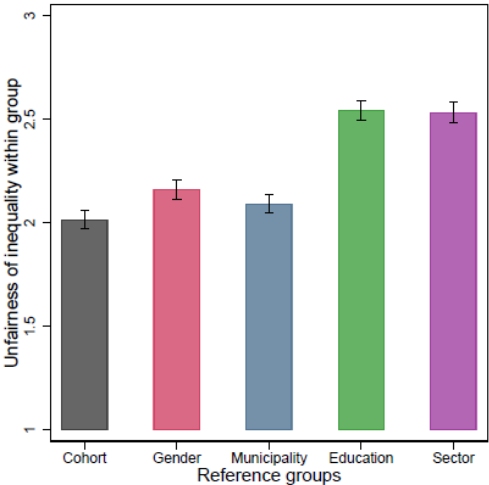
(C) UNFAIRNESS BY COHORT POSITION



(D) IMPORTANCE OF EFFORT



(A) UNFAIRNESS OF INEQUALITY



- Taxation of the Factor Capital
- Theory
 - Evidence
 - Examples from Austria

